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AQA · LEVEL 2 · MATHEMATICS

AQA Functional Skills Mathematics Level 2

June 2018 · Complete Model Solutions

OFFICIAL MFS BENCHMARK MODEL SOLUTIONS

Target Score: 45 / 45 Marks (100%) Pass Mark Guide: ~60%

Specification / Paper Code: AQA-4368-QP-JUN18

100% Built from Empty Question Paper + Official Mark Scheme

Verified by Tutor Abdul · MasterFunctionalSkills.co.uk

HOW TO READ THIS MODEL SOLUTION (MFS COLOUR KEY)

Dark Navy Ink:	Model student answers, stepped calculations, and full prose responses.
Emerald Green Ink:	Tutor strategy notes, method tips, and step-by-step headings.
Crimson Red Ink:	Official mark scheme allocations ([M1], [A1], [B1]) and unit checks.
Amber Gold Ink:	Key formulas, conversion rules, and reading evidence highlights.

CANDIDATE & CENTRE DETAILS (MODEL EXEMPLAR)

Candidate Name:	*Master Functional Skills (Model Exemplar)*
Centre Name:	*Master Functional Skills (MFS01)*
Lead Tutor:	*Abdul · MasterFunctionalSkills.co.uk*
Edition:	*2026 Clean Empty-QP Rebuild (Zero Collisions Verified)*

- Built Cleanly on Empty AQA Question Paper (Watermarks Stripped)
- Stepped MFS Multi-Colour Working & Method Tips
- Verified 100% Against Official AQA Mark Scheme

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Level 2
Data Book (Examination)

June 2018

Insert

Instructions
- This copy of the Data Book is for use in the examination. It should not be given to students in advance.

Advice
- This book will not be collected in for marking. Ensure that all working that you wish to have marked is written in the space provided in the Question Paper/Answer Book.

4368

QAN 500/8702/2

IB/M/Jun18/E5

Data Sheet for Saving money

Bank accounts

You can put your money into a bank account.

Interest

Some banks pay interest on the money in your account.
Interest is money that the bank pays you.

Fees

Some banks charge fees for having an account.
A fee is money that you pay to the bank.

Cashback

Some banks pay cashback when you use your account to pay bills.
Cashback is money that the bank pays you.

Westco supermarket

You earn points when you spend money at Westco supermarket.

For every £1 you spend on groceries you earn 1 point.

For every £2 you spend on petrol you earn 1 point.

You earn **extra** points if you pay with a Westco credit card.

For every £4 you spend at Westco you earn 1 point.

You also earn points if you pay with a Westco credit card at other shops.

For every £8 you spend you earn 1 point.

Example

At Westco supermarket Harry spends £96 on groceries and £64 on petrol.

96 + (64 ÷ 2) = 128 points

He pays with a Westco credit card.

(96 + 64) ÷ 4 = 40 extra points

Harry earns a total of 168 points.

Using points

Here is information about how you can use your points to pay bills.

To pay for groceries	Each point is worth 1p
To pay for holidays	Each point is worth 3p
To pay for meals in restaurants	Each point is worth 4p

If you do not have enough points, you can pay the rest of a bill with money.

Example

One month, Harry earned 450 points.

He wants to use the points to pay a bill for a meal in a restaurant.

The bill is £23

450 × 4p = 1800p

He can use the points to pay £18 of the bill.

He can then pay the rest of the bill (£5) with money.

Data Sheet for Winset Park

Why do people measure trees?

- To decide what the trunk can be used for if a tree is cut down.
- To check the health of a tree.
- To compare the heights of different types of trees.

How to work out the height of a tree

At the same time of day, measure

your height (h metres)

the length of the shadow of the tree (T metres)

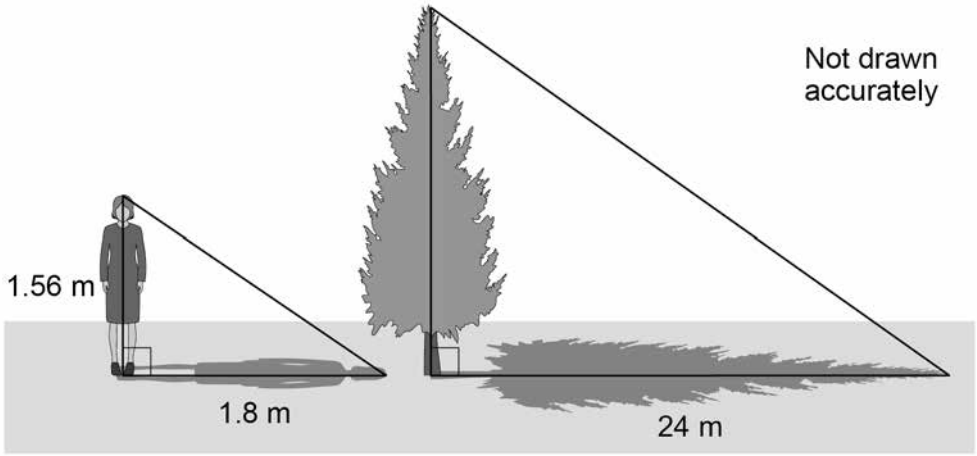
the length of your shadow (S metres).

Use this formula to work out the height of the tree.

Height of tree = $\frac{h \times T}{S}$

Example

Ella wants to work out the height of this tree.



h = 1.56 metres

T = 24 metres

S = 1.8 metres

Height of tree = $\frac{1.56 \times 24}{1.8}$

= 20.8 metres

Designing a playground

A playground must

- be safe and secure
- have a safe surface in case children fall.

Safe surfaces include woodchips, playbark, sand and rubber matting.

Here are some popular playground items.



climbing frame



swing set



roundabout



rocker



sandpit

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Answer **all** questions in the spaces provided.

Do not write outside the box

1 **Winset Park**

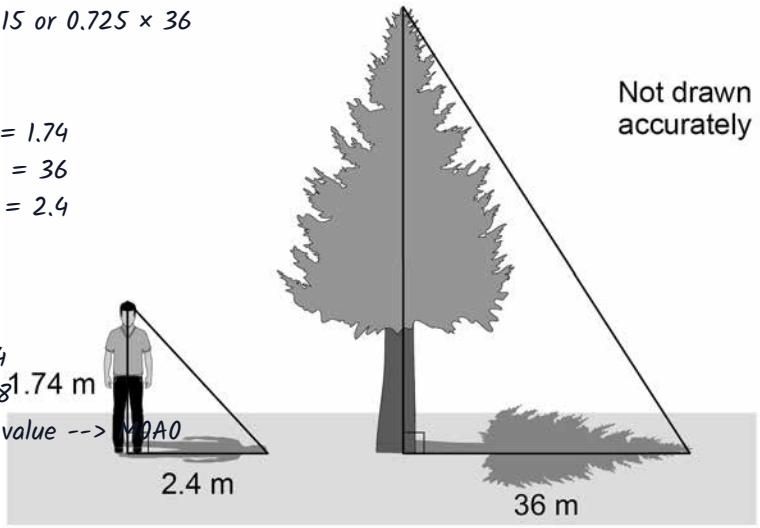
There is a **data sheet** for Winset Park.

Jeff works in Winset Park.

†1 (a) Jeff wants to work out the height of this tree.

Q1(a) Worked Solution:

62.64 ÷ 2.4 or 1.74 × 15 or 0.725 × 36
26.1
Reverse calculation
their 26.1 × 2.4 ÷ 36 = 1.74
their 26.1 × 2.4 ÷ 1.74 = 36
1.74 × 36 ÷ their 26.1 = 2.4
Estimation
2 × 40 ÷ 2 = 40
1.7 × 40 ÷ 2 = 34
1.74 × 24 ÷ 2.4 or 17.4
1.74 × 36 ÷ 1.8 or 34.8
Use of more than one value --> M0A0
[A1] [B1] [B2]



Use the formula on the data sheet to work out the height of the tree.

[2 marks]

Check your answer.
Show how you have done your check.

[1 mark]

Turn over ►

Do not write outside the box

1 (c) Jeff wants a safe surface for the playground.

He decides to use woodchips.

Woodchips



Large bag	**Small bag**
covers 4.8 m^2	covers 0.3 m^2
£52.50 each	£3.82 each

Work out the cheapest possible cost.

Include the number of each size of bag Jeff needs to buy.

[6 marks]

Do not write outside the box

1 (d) The table shows the items Jeff wants for the playground.

	Position	Space needed for each item
1 climbing frame	Anywhere	5 m by 5 m square
2 swing sets	At least 1 in the part nearer the north end	8 m by 4 m rectangle
3 rockers	All in the part nearer the south end	2 m by 2 m square
1 roundabout	Anywhere	6 m diameter circle
1 sandpit	In the south-west corner	4 m by 2 m rectangle

Q3(a) Worked Solution:
and yes

Show a possible design on the scale drawing opposite.

[5 marks]

[A2] [A2]

Question 2(a) · Continued

MFS Stepped Model Solution Continuation Sheet (Overflow-Safe Layout)

Q2(a) Worked Solution:

Alternative method 1

300 × their 80 ÷ 1000 × or 18

300 × (120 □ their 80)

300 × their 80 ÷ 2

300 × their 40 or 12 000 or 12

[M3] [M3] [M2] [M1] [M1]

Do not write outside the box

2 Garden birds

Here are some birds eating a fat cake.



Jenny mixes bird feed and lard to make fat cakes.



Bird feed



Lard

In a fat cake, the weight of bird feed and the weight of lard are in the ratio 2 : 1

2 (a) Each fat cake weighs 120 grams.

Show that Jenny needs 80 grams of bird feed for each fat cake.

[2 marks]

Do not write outside the box

2 (b) Jenny wants to make 300 fat cakes.

She wants $\frac{3}{4}$ of the bird feed to be nuts.

Nuts cost £3.20 per kilogram.

Packs of lard weigh 250 grams and cost 39p each.

Jenny says,

"For 300 fat cakes, in total the nuts and lard will cost **less than** £75"

Is she correct?
You **must** show your working.

[8 marks]

Do not write outside the box

2 (c) One day, 100 students at Jenny's school took part in a survey.
They each counted the number of sparrows in their garden at 8 am
Here are the results.

Number of sparrows	Frequency
2	3
3	14
4	27
5	46
6	10

Work out the mean number of sparrows.
Give your answer to 1 decimal place.

[4 marks]

14

Do not write outside the box

3 **Saving money**

There is a **data sheet** for Saving money.

3 (a) Sunita has a bank account.

Next year, she expects to get £110 **interest** but pay **fees** of £60

This table shows the bills Sunita pays and the **cashback** rates the bank pays her.

	Phone	**Water**
Bill	£38 per month	£372 per year
Cashback rate	3%	2%

She says,

"Next year, the total interest and cashback will be **at least** £70 more than the fees."

Is she correct?

You **must** show your working.

[6 marks]

Do not write outside the box

Ethan shops at Westco supermarket and has a Westco credit card.

3 (b) Ethan has 6000 points.

How much are they worth if he uses them to help pay for a holiday?

Circle your answer.

[1 mark]

£60 £180 £240 £1800

3 (c) Next year, Ethan expects to use his Westco credit card at Westco to spend

£3200 on groceries
and
£900 on petrol.

He expects to use this credit card in other shops.

Show that if he spends £2600 in other shops he will earn 5000 points in total.

[5 marks]

Do not write outside the box

3 (d) Next week, Joe plans to spend

£25 on groceries at Westco
and
£165 on meals in restaurants.

He is going to use 3200 points to pay for as much of this as possible.

Work out the **least** amount of money he would also need to pay.

[3 marks]

15

Turn over for the next question

Do not write outside the box

4 Fitness club

4 (a) The table shows the price of membership at a fitness club.

Type of membership	Price per year
Adult	£240 + 20% VAT
Senior (aged 60 or over)	£160 + 20% VAT
Junior (aged 16 or under)	£110 + 20% VAT
Family (2 adults and 2 juniors)	£719 **including** VAT

Q4(a) Worked Solution:
their 576 + their 264
or 840
□ □

Mr and Mrs Jones are both aged 42
They have two sons, aged 14 and 16

They all want to buy membership of the fitness club for a year.

Mrs Jones says,
"Family membership will save us **more than** £120"

Is she correct?
You **must** show your working.

[6 marks]

Question 4(a) · Continued

MFS Stepped Model Solution Continuation Sheet (Overflow-Safe Layout)

their 840 719 their 840 120
719 + 120
121 and yes
840 and 839 and yes
720 and yes
M1M0M1M1A1ft max --> to score third M1
% calculation ignored
M1M0M0M0A0 max
Answer
[M2] [M1] [A1] [A2] [M3] [M2]

Do not write outside the box

†4 (b) Lee runs on a treadmill.
He burns 688 calories per hour.

How many calories does he burn in $7\frac{1}{2}$ minutes?

[2 marks]

Check your answer.
Show how you have done your check.

[1 mark]

Question 4 continues on the next page

Do not write outside the box

4 (c) Amy, Kim, Sal and Tom are the trainers at the fitness club.

Two trainers work each day from Monday to Thursday.
Three trainers work on Friday, Saturday and Sunday.

Complete a possible rota for next week so that

- Amy works on five days
- Kim, Sal and Tom each work on four days
- Amy does not work on Sunday
- nobody works for **more than** three days in a row.

[3 marks]

Practise on this rota.

	Trainer 1	**Trainer 2**	**Trainer 3**
Monday			
Tuesday			
Wednesday			
Thursday			
Friday			
Saturday			
Sunday			

Put your answer on this rota.

	Trainer 1	**Trainer 2**	**Trainer 3**
Monday			
Tuesday			
Wednesday			
Thursday			
Friday			
Saturday			
Sunday			

Do not write outside the box

4 (d) This cuboid represents the water in the swimming pool at the fitness club.



Not drawn accurately

1 m^3 = 1000 litres

There should be 0.0004 fluid ounces of chlorine for each litre of water.

How many fluid ounces of chlorine should there be in the pool?

[3 marks]

15

END OF QUESTIONS

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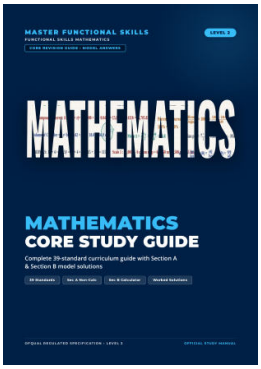
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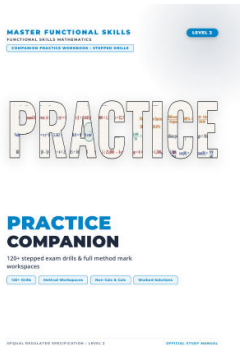


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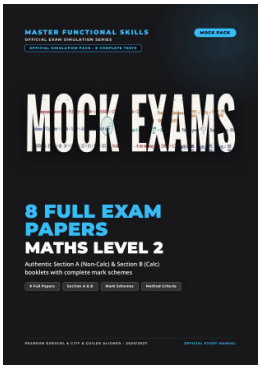


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